

HAT-P-18b

R-2092

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_260321 · created 2026-08-10T01:00:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461120.8618 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.097 +/- 0.047
Transit depth (Rp/Rs)^2	0.94 +/- 0.92 [%]
Orbital Inclination (inc)	89.34 +/- 1.97
Airmass coefficient 1 (a1)	0.996 +/- 0.014
Airmass coefficient 2 (a2)	-0.001 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.41 %
Best Comparison Star	#2 - [471, 328]
Optimal Method	PSF photometry
Transit Duration (day)	0.072 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.097 ± 0.047 vs 0.1356 (deviation 0.82σ)
Duration fit vs published	0.072 d vs 0.1129 d (deviation 1.95σ)
Mid-time O-C	1.3 min
Depth SNR	2.1
Residual scatter	1.41 %

Light curve

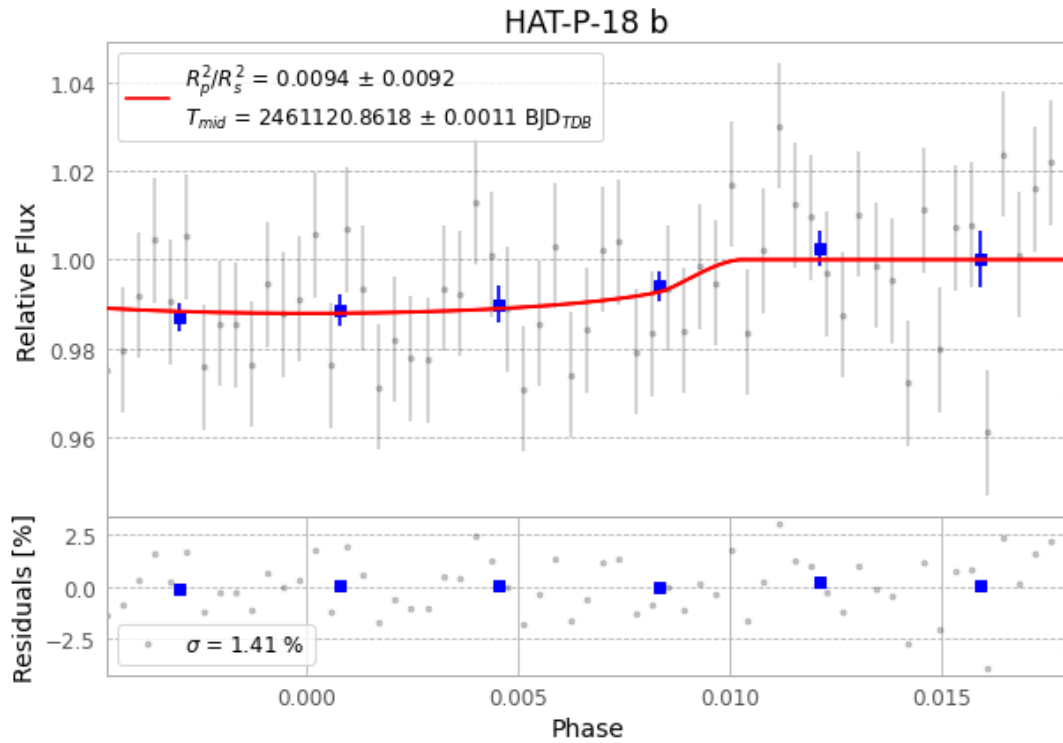


Figure 1 — FinalLightCurve_HAT-P-18 b_2026-03-21.png (SHA-256 393fe49b8f934bcc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [233, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 598b833be56ef470...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

61 FITS frames (40 MB), HATP-18260321080015.FITS → HATP-18260321110015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-260321.log (d7b10239b2e3...), FinalLightCurve_HAT-P-18 b_2026-03-21.png (393fe49b8f93...), FinalLightCurve_HAT-P-18 b_2026-03-21.csv (7c6f929d8e6e...)

Compute resources

Wall time 141.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 01:00Z.